Order Flow Imbalance and the Micro-Price

What moves the price at short horizons, and the best guess of where it goes

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The question

Over seconds, prices move because queues at the best quotes are added to and eaten. **Which summary of order flow explains the move, and which summary of the book predicts the next one?**

- **Contemporaneous**: order flow imbalance (OFI) explains price changes in the same window.
- **Predictive**: queue imbalance forecasts the next move; the micro-price turns it into a fair value.

An LP quotes around a fair value. Quoting around the mid when the micro-price is elsewhere hands edge to the next trader.

In this deck

1. OFI: definition, linear impact, depth.
2. Multi-level and integrated OFI.
3. The micro-price and how to estimate it.
4. Evidence, and the statistics of overlapping windows.

Order flow imbalance

From consecutive best-quote snapshots $n-1 \to n$, with best bid and ask prices p_B, p_A and sizes q_b, q_a (Cont et al., 2014):

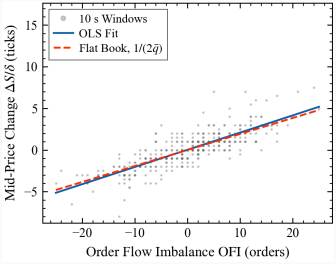
$$e_n = \mathbf{1}_{\{p_{B,n} \geq p_{B,n-1}\}} q_{b,n} - \mathbf{1}_{\{p_{B,n} \leq p_{B,n-1}\}} q_{b,n-1} - \mathbf{1}_{\{p_{A,n} \leq p_{A,n-1}\}} q_{a,n} + \mathbf{1}_{\{p_{A,n} \geq p_{A,n-1}\}} q_{a,n-1}.$$

New bids, cancelled asks and market buys push e_n up. Summed over a window of length Δt (s), $\mathrm{OFI} = \sum e_n$ (shares or orders).

Stylized model. If every level holds $\bar{q}$ orders, OFI moves one quote by $\mathrm{OFI}/\bar{q}$ ticks and the mid by half as much:

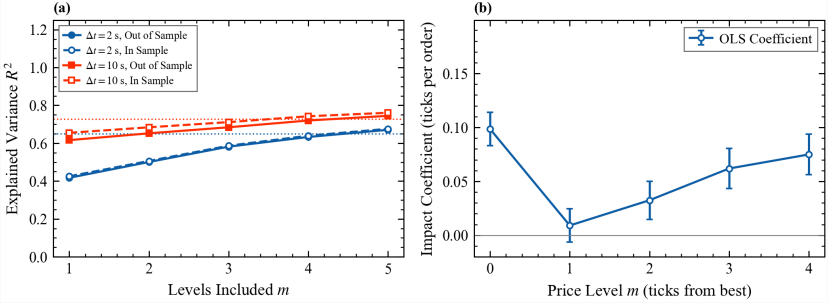
$$\Delta S/\delta = \kappa\,\mathrm{OFI} + \xi, \qquad \kappa \approx \frac{1}{2\bar{q}},$$

with mid change ΔS (currency), tick δ and residual ξ.



Markovian book: $\kappa = 0.2067 \pm 0.0072$ against $1/(2\bar{q}) = 0.1921$; $R^2 = 0.66$, 412 windows of 10 s.

Deeper levels: multi-level and integrated OFI



Multi-level OFI: the same formula at each level m ticks behind the best (Xu et al., 2019). **Integrated OFI**: level OFIs divided by mean depth and combined with the first principal component weights (Cont et al., 2023). Here the weights are $0.22, 0.23, 0.20, 0.19, 0.16$.

Out-of-sample R^2, $\Delta t = 10$ s (823 windows): best level 0.62, 5 levels 0.74, integrated 0.73. At $\Delta t = 2$ s: 0.42, 0.67, 0.65. Once the best queue empties, the next level decides how far the price moves.

Statistics of OFI regressions

Tautology. OFI includes the events that move the price, so part of R^2 is mechanical. Cont et al. (2014) re-estimate without price-changing events: R^2 falls from about 65% to the 35% to 60% range, still large.

Heteroskedasticity. Residual variance follows depth and time of day. Use White (heteroskedasticity-consistent) errors; for regressions of estimated coefficients across windows, Newey-West errors.

Nonstationary coefficient. κ varies with depth inside the day, so estimate it on short subsamples (CKS: half an hour) or model $\kappa \propto \bar{q}^{-1}$.

Overlapping horizons. Forecast regressions of the mid change over h on today's imbalance, with one sample per event, have strongly overlapping targets. Effective sample size is closer to T/h than to the number of events. Block bootstrap or Hansen-Hodrick errors.

Out of sample. Fit on the first half, test on the second; with many levels in-sample R^2 always rises.

The micro-price

State $x = (I, s)$: imbalance $I = q_b/(q_b + q_a)$ in buckets and spread s (ticks). The micro-price (Stoikov, 2018) is the limit of the expected mid after the next k mid changes:

$$P^{\text{micro}} = M + g_1(x) + g_2(x) + \cdots, \qquad g_{i+1} = B\, g_i,$$

with M the mid (ticks). From consecutive observations:

$$g_1 = (I_d - Q)^{-1} R\, K, \qquad B = (I_d - Q)^{-1} T,$$

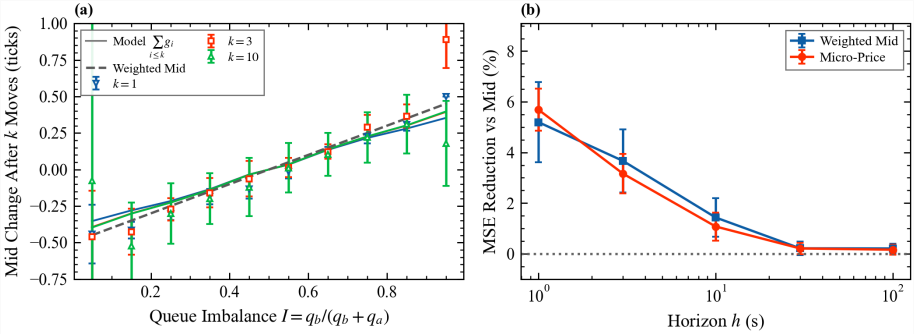
where Q holds transitions that keep the mid, T those that change it, $R(x, k)$ the probability of a mid change of k ticks, K the vector of changes and I_d the identity.

Symmetrize. Each observation also enters mirrored ($I \to 1 - I$, $\Delta M \to -\Delta M$). Then g_i is odd in imbalance and the series converges.

Compare with the weighted mid
$Ip_A + (1 - I)p_B = M + (I - \frac{1}{2})s$, which has no justification beyond intuition and is not a martingale.

Assumption. (I, s) is Markov. It is not exactly: queue sizes, not only their ratio, matter (deck 2).

The micro-price on a simulated book



Power-law book, 600,000 events; first half trains, second half tests. (a) Model $\sum_{i\le k} g_i$ against realized mid changes after k moves, block errors: in inner buckets the worst gap is 4.7 s.e., a sign that (I, s) is not quite Markov. G reaches 0.40 ticks at one-tick spread.

(b) Mean squared error of the future mid relative to the mid: micro-price $5.7 \pm 0.4\%$ better at 1 s, weighted mid $5.2 \pm 0.8\%$; at 10 s, $1.1 \pm 0.3\%$ and $1.4 \pm 0.4\%$. In this book $G(I)$ is nearly linear, so the two are close; real books differ (next slide).

What the data show

- **OFI explains short-horizon moves.** Fifty S&P 500 stocks drawn at random, TAQ data for April 2010, 10-second windows, half-hour regressions: average R^2 of 65%. A quadratic term raises it only to 68%. The slope is inversely proportional to depth; near the open depth is about half its average and impact about twice its average (Cont et al., 2014).
- **Trades alone explain less.** The relation between price changes and trade volume is noisier and less robust than the one with OFI (Cont et al., 2014).
- **Deeper levels help.** Multi-level OFI improves the fit over the best level (Xu et al., 2019). Integrated OFI makes contemporaneous cross-impact between stocks redundant, while lagged cross-asset OFI forecasts short-horizon returns (Cont et al., 2023).
- **Micro-price.** Estimated on high-frequency data, it predicts short-term prices better than the mid or the weighted mid (Stoikov, 2018).
- **Deep models.** Networks on raw book states forecast short-term moves out of sample (Zhang et al., 2019); deep OFI features forecast returns at several horizons (Kolm et al., 2023); pooled training finds a universal price formation mechanism (Sirignano and Cont, 2019).

Discussion and takeaways

For a liquidity provider

- Quote around the micro-price, not the mid. The gap is up to a few tenths of a tick at the extremes of imbalance here, comparable to half the spread.
- OFI is contemporaneous; imbalance is predictive. A strategy can only use the second in real time.
- Imbalance is a signal with a finite life. How much of it to put into quotes depends on that life (deck 7).

Open questions

- A micro-price that uses queue sizes and recent flow, with uncertainty, and stays a martingale.
- OFI across fragmented venues with latency between them.
- Whether generative models of the book reproduce the OFI slope and its depth dependence (frontier part of the main deck).

Reproduce

Code: `impact.py` (OFI, multi-level, integrated), `microprice.py`. Experiments: `e8_ofi.py`, `e14_mlofi.py`, `e20_microprice.py`.

```
from stochastic_lob import microprice as mp
idx, imb = mp.states(rec.q_bid, rec.q_ask, rec.best_ask - rec.best_bid)
fit = mp.fit(idx[:n // 2], mid[:n // 2])          # g_1, B, G per state
mid_, wmid, micro = mp.predictors(idx, imb, mid, spread, fit["G"])
```

References i

Rama Cont, Arseniy Kukanov, and Sasha Stoikov. The price impact of order book events. *Journal of Financial Econometrics*, 12(1):47–88, 2014.

Rama Cont, Mihai Cucuringu, and Chao Zhang. Cross-impact of order flow imbalance in equity markets. *Quantitative Finance*, 23(10):1373–1393, 2023.

Petter N. Kolm, Jeremy Turiel, and Nicholas Westray. Deep order flow imbalance: Extracting alpha at multiple horizons from the limit order book. *Mathematical Finance*, 33(4): 1044–1081, 2023.

Justin Sirignano and Rama Cont. Universal features of price formation in financial markets: perspectives from deep learning. *Quantitative Finance*, 19(9):1449–1459, 2019.

Sasha Stoikov. The micro-price: a high-frequency estimator of future prices. *Quantitative Finance*, 18(12):1959–1966, 2018.

Ke Xu, Martin D. Gould, and Sam D. Howison. Multi-level order-flow imbalance in a limit order book, 2019. Market Microstructure and Liquidity; arXiv:1907.06230.

Zihao Zhang, Stefan Zohren, and Stephen Roberts. DeepLOB: Deep convolutional neural networks for limit order books, 2019. arXiv:1808.03668.